Nonlinear Resonance Characterization of Parametrically Excited Pre-buckled Elastic Struts: Supplementary Material

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S1 Directly excited case: Nonlinear resonance characteristics

Presently, the directly excited nonlinear system is considered thus:

$$\ddot{\tilde{Q}} + \tilde{c}\dot{\tilde{Q}} + \alpha_1\big(1-\tilde{P}_0\big)\tilde{Q} + \alpha_i\tilde{Q}\big(\dot{\tilde{Q}}^2 + \tilde{Q}\ddot{\tilde{Q}}\big) + \big(\alpha_3 - \alpha_{3p}\tilde{P}_0\big)\tilde{Q}^3 = \tilde{f}_{\mathrm{t}}\cos\left(\pi^2\sqrt{1-\tilde{P}_0}\tilde{\Omega}\tilde{t}\right), \tag{S1}$$

where:

$$\alpha_1 = \pi^4, \quad \alpha_i = \frac{\pi^2}{48}\left(8\pi^2 - 9\right), \quad \alpha_3 = \frac{\pi^6}{2}, \tag{S2}$$

$$\alpha_{3p} = \frac{3\pi^6}{8}, \quad \alpha_R = \frac{1}{4}, \quad \xi = \frac{\tilde{c}}{2\pi^2\sqrt{1-\tilde{P}_0}}. \tag{S3}$$

and all the symbols are defined in the main body of the article.

Eq. (S1) is solved following the same procedure described in Section 3 of the article, using the same damping ratio $\xi = 2\%$. The frequency–response functions for prescribed values of $\tilde{P}_0$ and increasing excitation amplitude $\tilde{f}_{\mathrm{t}}$, together with the corresponding backbone curves at the same $\tilde{P}_0$, are summarised in Figure S1.

The results show that, for small values of $\tilde{P}_0$ (e.g. $\tilde{P}_0 = 0.1, 0.3, 0.5$), the resonance peaks exhibit a clear leftward shift, following the softening-type 'backbone' curve. As $\tilde{P}_0$ approaches the transition value $\tilde{P}_{0,\mathrm{n}}$ (e.g. $\tilde{P}_0 = 0.55, 0.6, 0.65$), the softening behaviour weakens and becomes less pronounced compared with the 'backbone' curve. A weak hardening tendency is observed at $\tilde{P}_0 = 0.7$, indicating that the transition value of the system is approximately $\tilde{P}_0 = 0.7$, which is lower than the 'backbone-based' estimation, i.e., $\tilde{P}_{0,\mathrm{n}} = 0.79$. This discrepancy is attributed to the approximation of the 'backbone' curve using a single-mode harmonic balance truncated at cubic order. A more accurate transition value is expected when higher harmonics and higher-order nonlinear terms are included for the 'backbone' curve. For values of $\tilde{P}_0$ exceeding the transition value (e.g. $\tilde{P}_0 = 0.8, 0.9$), the resonance peaks follow the hardening-type 'backbone' curve. Overall, for the directly excited system, the excitation amplitude $\tilde{f}_{\mathrm{t}}$ has a relatively minor influence on the qualitative characteristics of the resonance peak, and the 'backbone' curve provides a reliable predictor of the resonance behaviour.

S2 Characteristics of the PPR peak: FE verification

The FE verification is performed following the settings and procedures described in Section 3 of the article. The results are presented in Figures S2–S6.

A response is considered to have reached steady-state if, over the last ten excitation periods, more than three fully developed periodic cycles with visually identical trajectories are observed in both the time histories and phase portraits. The response magnitude is then taken as the maximum amplitude of these periodic cycles. For some loading scenarios, the response amplitude becomes excessively large ($\tilde{Q} > 0.3$), leading to convergence difficulties and preventing the development of three periodic

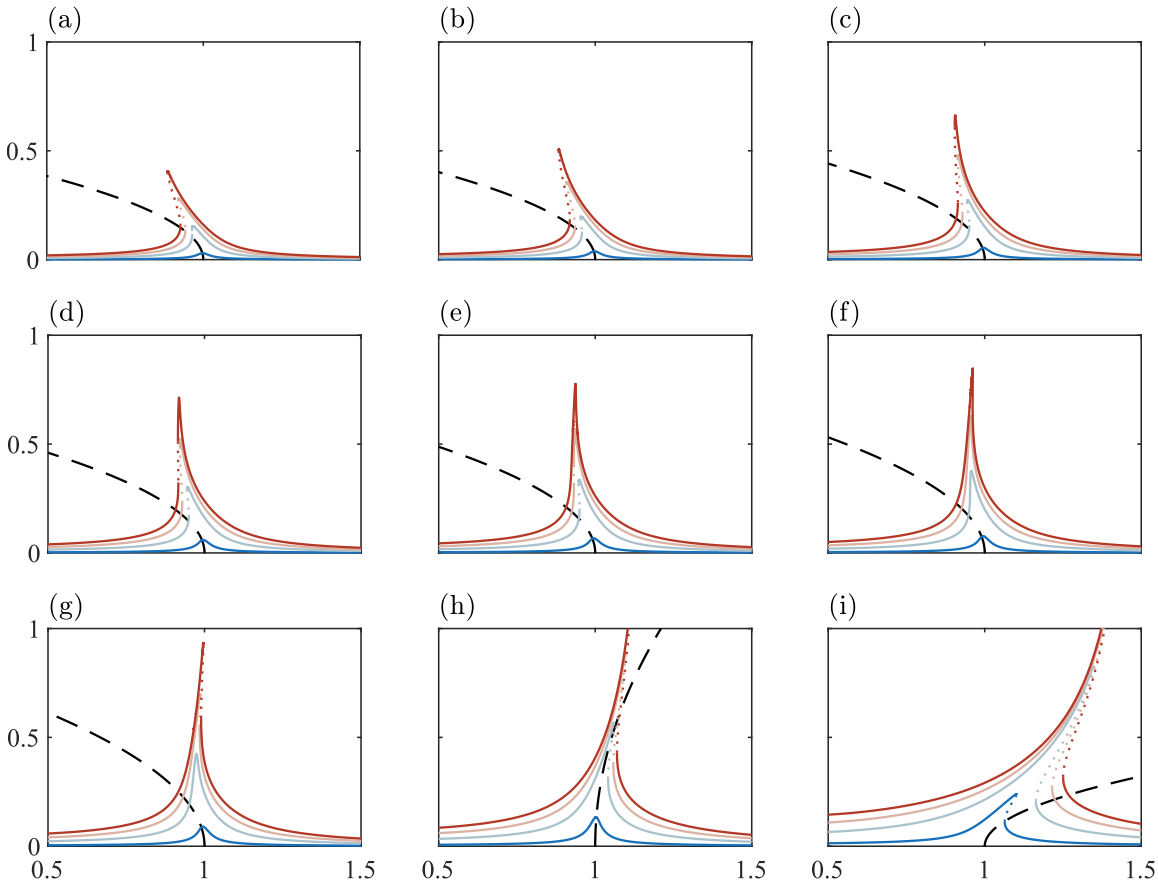

Figure S1: Frequency–response function of the directly excited system given in Eq. S1. Different colors, from blue to light blue, light red and red, denote different $\tilde{f}_{\mathrm{t}} = 0.1$, 0.5, 0.9, 1.3, respectively. The ordinate and abscissa correspond to $\tilde{Q}$ and $\tilde{\Omega}$, respectively. Black–dashed line and dotted lines denote the 'backbone' curve and unstable solution, respectively. (a) $\tilde{P}_0 = 0.1$; (b) $\tilde{P}_0 = 0.3$; (c) $\tilde{P}_0 = 0.5$; (d) $\tilde{P}_0 = 0.55$; (e) $\tilde{P}_0 = 0.6$; (f) $\tilde{P}_0 = 0.65$; (g) $\tilde{P}_0 = 0.7$; (h) $\tilde{P}_0 = 0.8$; (i) $\tilde{P}_0 = 0.9$.

cycles. Such cases are interpreted as large-amplitude responses, and the steady-state magnitude is approximated by the maximum amplitude over the last three excitation periods.

Frequency sweeping is implemented by using the response magnitude of the previous run as the initial displacement for the subsequent run. For given values of $\tilde{P}_0$ and $\tilde{P}_\mathrm{t}$, forward and backward frequency sweeps correspond to increasing and decreasing $\tilde{\Omega}$, respectively.

The characteristics of the resonance peak are identified by examining the presence of significant amplitude differences—typically of the order of 10^3—between forward and backward sweep responses under the same loading scenario. If such differences occur over a frequency range on the lower frequency side of the resonance peak, the response is classified as softening-type. Conversely, the presence of this structure on the higher frequency side indicates a hardening-type response. The absence of such hysteresis behaviour corresponds to a neutral-type resonance peak.

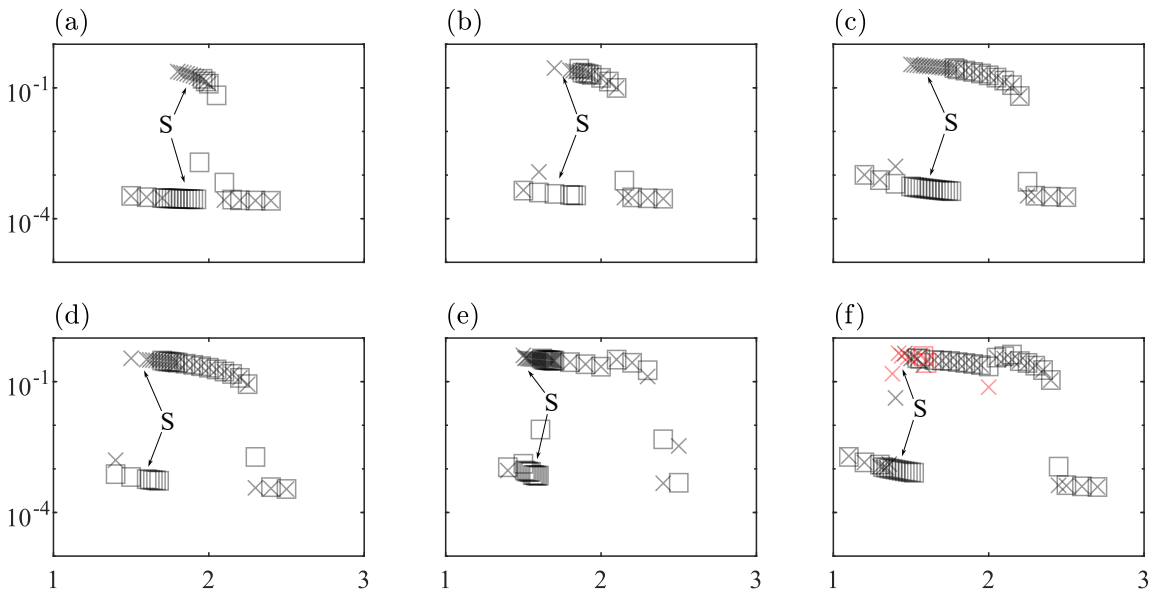

Figure S2: FE results for $\tilde{P}_0 = -0.3$. The ordinate and abscissa correspond to $\tilde{Q}$ and $\tilde{\Omega}$, respectively. Square and cross symbols denote frequency-increasing (forward-sweep) and frequency-decreasing (backward-sweep) FE results, respectively. Data points highlighting the response characteristics are indicated by arrows, with 'S' and 'H' denoting softening-type and hardening-type behaviour, respectively. Red symbols indicate responses with severe amplitudes for which steady-state solutions could not be sustained for more than three cycles, whereas black symbols correspond to responses that persist for more than three cycles. Panels (a)–(f) correspond to $\tilde{P}_\mathrm{t} = 0.2$, 0.4, 0.6, 0.8, 1.0, and 1.2, respectively; all cases exhibit softening-type resonance behaviour.

S3 Poincaré map and bifurcation diagram

The Poincaré maps and bifurcation diagram are presented in Fig. S7. Owing to the softening behaviour, two stable orbits coexist at $\tilde{\Omega} = 1.75$: a lower-amplitude period-1 orbit and a higher-amplitude period-2 orbit. At $\tilde{\Omega} = 2$, only a single stable period-2 orbit exists.

The bifurcation diagram exhibits a single branch, corresponding to a single Poincaré section point and period-1 solutions, except within the interval bounded by the two period-doubling points at $\tilde{\Omega} = 1.786$ and 2.203. Within this range, two branches emerge, corresponding to two Poincaré section points and period-2 solutions. The right PD is supercritical, as it provides a continuous transition between the single- and double-branch solutions, whereas the left PD is subcritical, characterised by a discontinuous transition via a jump.

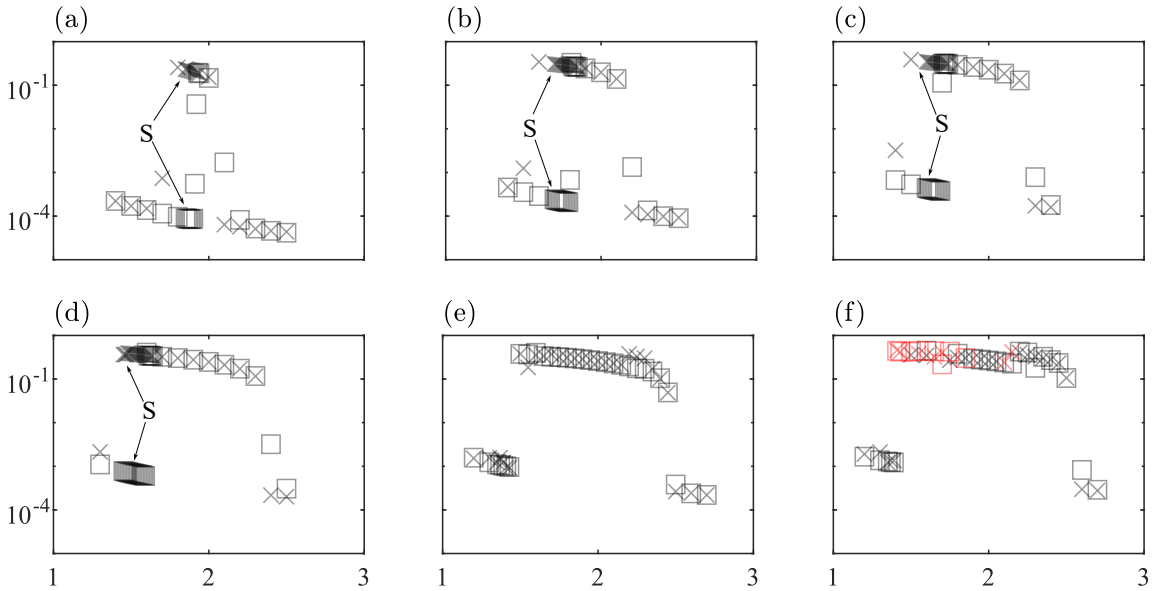

Figure S3: FE results for $\tilde{P}_0 = 0$. The symbols and notation are the same as those defined in Figure S2. Panels (a)–(d) correspond to $\tilde{P}_\mathrm{t} = 0.2$, 0.4, 0.6, and 0.8, respectively, and exhibit softening-type resonance behaviour; panels (e) and (f) correspond to $\tilde{P}_\mathrm{t} = 1$ and 1.3, respectively, and exhibit neutral-type resonance behaviour.

S4 Significance of axial inertia nonlinearity

Removing the nonlinear terms originated from axial inertia, Eq. (2.18) from the main body of the article becomes thus:

$$\ddot{\tilde{Q}} + \tilde{c}\dot{\tilde{Q}} + \alpha_1\big(1 - \tilde{P}\big)\tilde{Q} + \big(\alpha_3 - \alpha_{3p}\tilde{P}\big)\tilde{Q}^3 = \tilde{f}, \tag{S4}$$

where the coefficients are given in Eq. (2.20) of the main body of the article alongside the definition of all the symbols used. Presently, the system only has the Duffing-type nonlinearity originated from the large curvature.

As shown in Fig. S8, neglecting axial inertia fundamentally alters the resonance characteristics—both PPR and FPR—shifting the response from softening-type to hardening-type. This behaviour is inconsistent with the FE results, thereby highlighting the essential role of axial inertia in governing the dynamic response of the parametrically excited simply-supported strut.

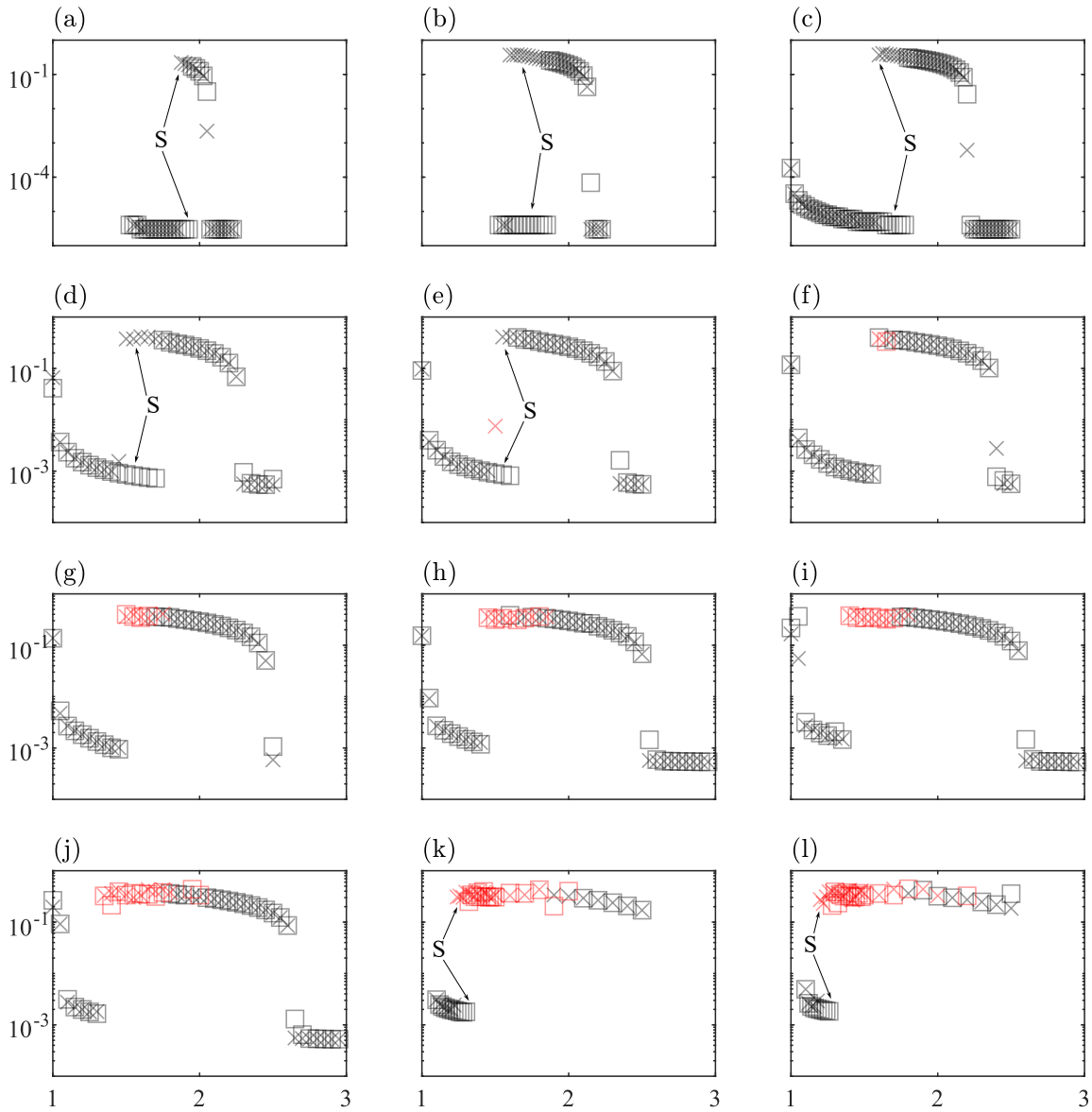

Figure S4: FE results for $\tilde{P}_0 = 0.3$. The symbols and notation are the same as those defined in Figure S2. Panels (a)–(e) correspond to $\tilde{P}_\mathrm{t} = 0.1$, 0.2, 0.3, 0.4, and 0.5, respectively, and exhibit softening-type resonance behaviour. Panels (f)–(j) correspond to $\tilde{P}_\mathrm{t} = 0.6$, 0.7, 0.8, 0.9, and 1.0, respectively, and exhibit neutral-type resonance behaviour. Panels (k) and (l) correspond to $\tilde{P}_\mathrm{t} = 1.1$ and 1.2, respectively, and exhibit softening-type resonance behaviour.

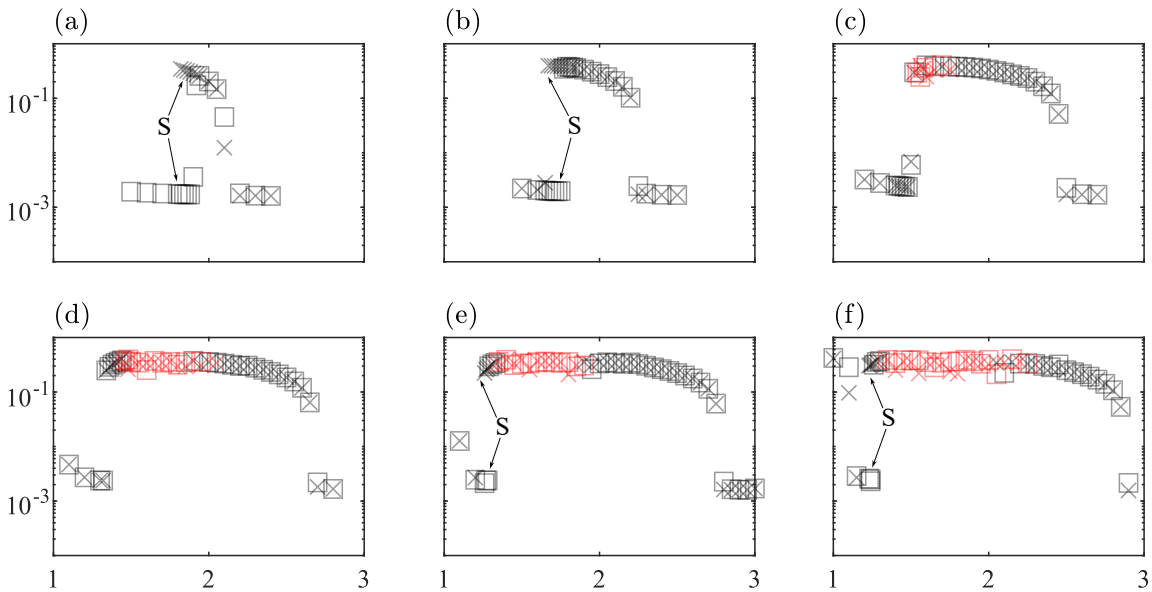

Figure S5: FE results for $\tilde{P}_0 = 0.6$. The symbols and notation are the same as those defined in Figure S2. Panels (a) and (b) correspond to $\tilde{P}_\mathrm{t} = 0.1$ and 0.2, respectively, and exhibit softening-type resonance behaviour. Panels (c) and (d) correspond to $\tilde{P}_\mathrm{t} = 0.4$ and 0.6 respectively, and exhibit neutral-type resonance behaviour. Panels (e)–(f) correspond to $\tilde{P}_\mathrm{t} = 0.7$ and 0.8, respectively, and exhibit softening-type resonance behaviour.

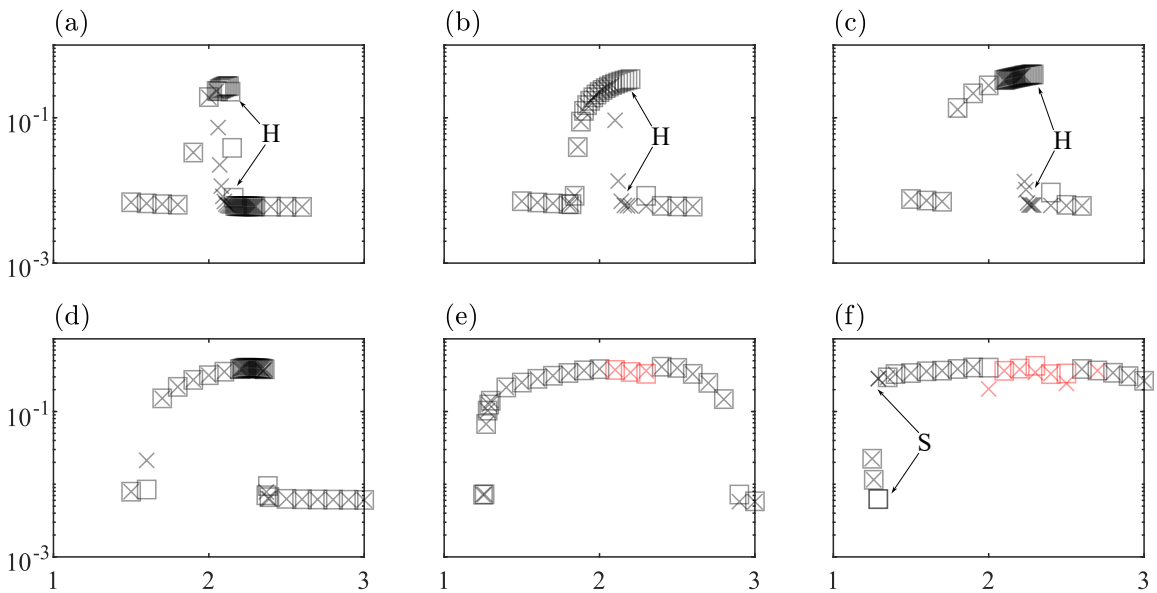

Figure S6: FE results for $\tilde{P}_0 \approx 1$. The symbols and notation are the same as those defined in Figure S2. Panels (a)–(c) correspond to $\tilde{P}_\mathrm{t} = 0.04$ while $\tilde{P}_0 = 0.85$, 0.9 and 0.95, respectively, and exhibit hardening-type resonance behaviour. Panels (d) and (e) correspond to $\tilde{P}_0 = 0.85$ while $\tilde{P}_\mathrm{t} = 0.12$ and 0.14, respectively, and exhibit neutral-type resonance behaviour. Panels (f) correspond to $\tilde{P}_0 = 0.85$ while $\tilde{P}_\mathrm{t} = 0.5$ and exhibit softening-type resonance behaviour.

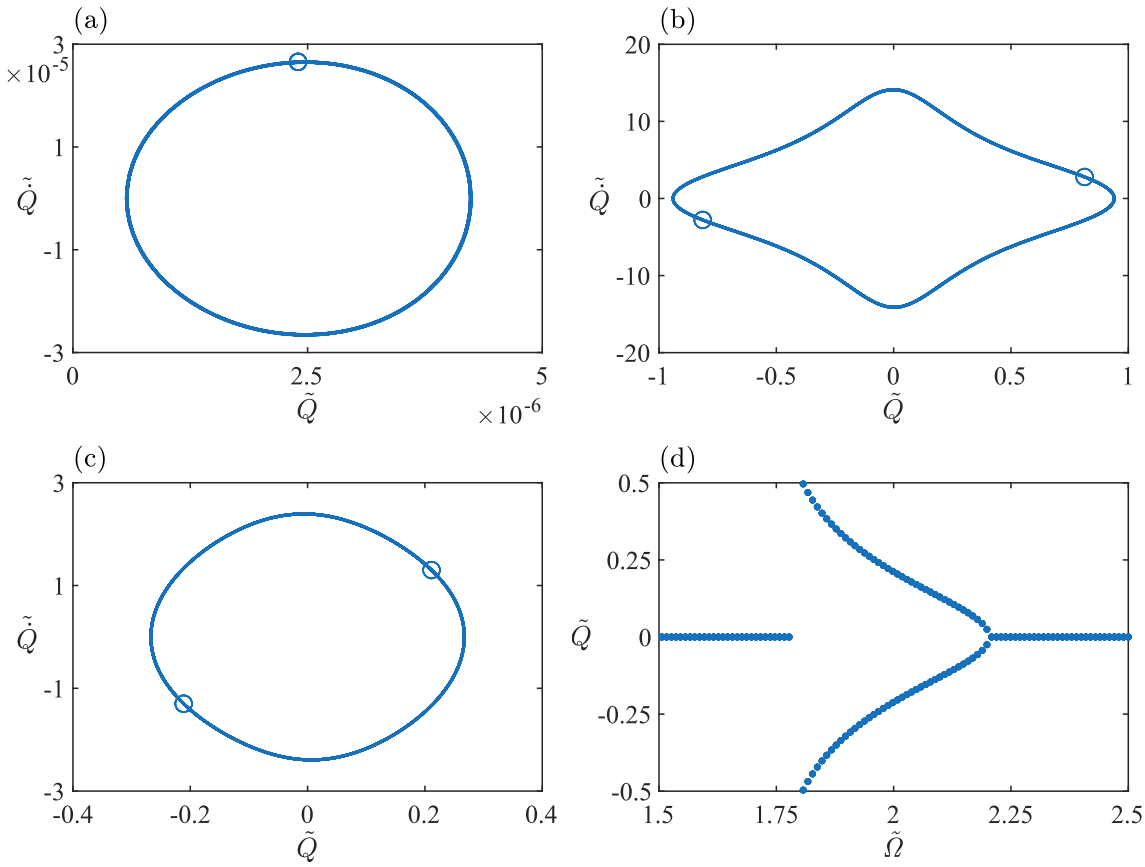

Figure S7: Poincaré maps and bifurcation diagrams of Eq. (2.18) for $\tilde{P}_0 = \tilde{P}_\mathrm{t} = 0.3$. In (a)–(c), blue hollow circles denote the Poincaré section points. (a) Lower-amplitude orbit at $\tilde{\Omega} = 1.75$; (b) higher-amplitude orbit at $\tilde{\Omega} = 1.75$; (c) orbit at $\tilde{\Omega} = 2$; (d) bifurcation diagram in the vicinity of the PPR peak.

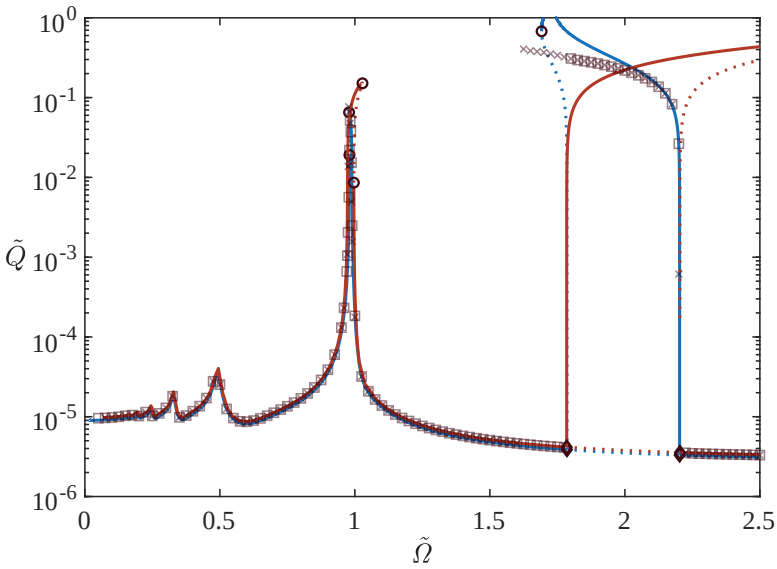

Figure S8: Comparison of frequency–response function from Eq. (2.18) (coloured blue) and Eq. (S4) (coloured red) with $\tilde{P}_0 = \tilde{P}_\mathrm{t} = 0.3$. Solid and dotted lines denote stable and unstable solutions respectively. Circle, diamond, cross and square symbols denote LPC, PD, frequency increasing (forward) sweeping FE results and frequency decreasing (backward) sweeping FE results, respectively.